

Portfolio Algorithms

INTRODUCTION

Portfolio algorithms and multi-period trade schedule optimization has gained momentum in the financial community due to the increase in program and algorithmic trading. By understanding how portfolio trading decisions influence returns, traders will be better prepared to make decisions consistent with the overall investment objectives of the fund. Unfortunately, traditional optimization techniques are not adequate for portfolio needs due to the non-linearity of the price impact function, the large number of decision variables, and the time it takes to calculate the answer.

Each time a trader is given a trade list to execute (e.g., basket, program, or portfolio) they face an inherent dilemma. Trading too quickly will result in too much price impact due to liquidity demands and information leakage but trading too slowly will result in too much risk which could lead to even higher costs in times of adverse price movement (“trader’s dilemma”).

In order to address the conflicting expressions of market impact cost and timing risk, traders derive a trade schedule (“slicing strategy”) that balances the trade-off between price impact and risk based on a specified level of risk aversion. The appropriate computational technique to solve this problem for a portfolio is the multi-period trade schedule optimization. But unlike the portfolio manager who usually has ample time to run sophisticated optimization algorithms and perform thorough sensitivity analysis, a timely solution for the trader is mandatory especially considering that many times they are given the order just before the market open or during the trading day. Most currently available optimization routines take too much time to solve the trader’s dilemma to be useful for investors. These packages can take several minutes, hours, or more, especially if the problem involves thousands of stocks over a long trading horizon. Traders require real-time solutions in seconds or less.

Trade schedule optimization to minimize total trading costs has been previously studied. For example, Bertsimas and Lo (1998) provided an approach to minimize price impact in the presence of expected future